Trusted Aunti

A safety-aware assistant for adolescent girls in Kenya: contraception, sexual health, staying safe, and getting to a service.

Build report — the refined version.

github.com/George-techie/girl-effect-srh-rag · 176 tests · 8 governed sources, 1,693 chunks · 8 verified services across 8 routes

What the brief asked for

The previous submission was assessed as *overengineered in places*, with a request to see what a more refined version focused on delivery of the use case would look like. The scope also narrowed: **sexual and reproductive health, centred on contraception and family planning**, plus the empowerment outcomes that follow. Mental health and menstruation left the topic list.

This build is the answer to both. Same solution philosophy. Every component either justified by a measurement or deleted. **Refined does not mean smaller** — two components were added during the work, each because something measurable failed without them.

$0.0109	52/52	1.000	24/24
COST PER TURN	ROUTING ACCURACY	SAFEGUARDING RECALL & PRECISION	GROUNDED ANSWERS CITING

Performance against the previous build

All previous figures below are read from that project's own recorded evaluation runs of 4 August 2026, 51 cases each, not reconstructed.

CONFIGURATION	ACC.	UNSAFE	UNHELPFUL	CALLS/TURN	TOKENS/TURN	COST/TURN	MEDIAN LATENCY
A — no safeguarding layer	0.784	7	1	0.98	3,938	–	5,476 ms
B — safeguarding, no evidence judge	0.882	0	5	2.65	3,848	–	7,058 ms
C — B plus the LLM evidence judge	0.745	0	12	3.53	6,838	–	7,030 ms
B+ — full profile, five model roles	0.843	1	6	3.78	8,599	$0.0189	12,410 ms
This build	52/52 †	–	3	0.69	4,946	$0.0109	3,536 ms

† Not the same measure. The previous action accuracy was scored by an LLM judge over 51 cases against a different scope and corpus; this build's 52/52 is deterministic routing accuracy. The columns that are directly comparable are calls, tokens, cost and latency — all measured over 51 and 52 cases respectively, priced with the same OpenRouter table.

On the refusal column: the previous build's "unhelpful actions" were scored by an LLM judge; this build's 3 are counted deterministically — blocked by the validator, no evidence found, or a provider error. Not the same instrument, so read it as the same order of magnitude rather than a precise ranking. The eight out-of-scope declines are excluded, because declining a question the service deliberately does not cover is correct behaviour rather than a refusal.

5.5× FEWER MODEL CALLS PER TURN	**1.7×** FEWER TOKENS PER TURN	**42%** LOWER COST PER TURN	**38%** OF TURNS NEED NO MODEL

What that comparison does and does not support

Cost and consumption are the robust numbers. They are properties of the architecture and do not move with network conditions. **0.69 model calls per turn against 3.78** — a 5.5× reduction — and **4,946 tokens per turn against 8,599**, which prices at **$0.0109 per turn against $0.0189**. Twenty of the fifty-two messages reached no model at all, and those are the turns where it matters most: every safeguarding reply is assembled from approved text and verified table rows in **0 ms**.

Per-turn call rate varies with the mix of questions. On the 52-message decision set, which is deliberately heavy in safeguarding and out-of-scope cases, it is 0.69. On the 38-turn conversation set, which is mostly questions, it is 1.03. Both are reported rather than the flattering one.

Latency is a weaker claim and is presented as one. The previous project's own two runs of the identical System B recorded 3,993 ms and 7,058 ms — a 3,065 ms spread from provider variance alone. Against B+ the improvement is large and consistent with removing most of the model calls; against B it is inside the noise. The defensible statement is that this build does the same work in roughly one call instead of four, and latency followed.

The two accuracy columns are not directly comparable: the previous action accuracy was scored by an LLM judge over 51 cases against a different scope and a different corpus. This build's equivalent is 52/52 on deterministic routing, measured differently and reported separately below.

Evaluation, by dimension

The previous build shipped **System D** — its full configuration plus a conversation layer. D itself was never scored, so the closest benchmarked configuration is **System C**. Blank cells are honest: that build did not measure the dimension.

	METRIC	SYSTEM C	THIS BUILD	WHAT IT TELLS US
Safeguarded	Safeguarding recall	1.000	12/12	known disclosures detected
	Safeguarding precision	–	12/12	no unnecessary escalation
Accurate	Retrieval Adequate@5	–	0.960	the evidence can actually answer her
	Grounded turns citing	–	24/24	factual answers stayed grounded
	Routing accuracy	0.766 F1	52/52	
Reliable	**Unusable outcomes**	12/51 · 23.5%	3/52 · 5.8%	turns where she gets nothing useful *
	Multi-turn path accuracy	–	21/23	context survives a conversation
	Forbidden retrievals	–	0/23	context stopped pulling wrong evidence
Resonant	Register match	–	38/38	answered in the language she wrote in
	Natural continuation	–	32/37 · 86%	the reply gives her somewhere to go
	Standing offers avoided	–	36/37	fewer dead-ending replies
	Deferrals avoided	–	38/38	never asks permission to answer what she asked
Cost	LLM calls per turn	3.53	0.69	
	Tokens per turn	6,838	4,946	
	Median latency	7,030 ms	3,536 ms	

** Not like-for-like. System C's unhelpful actions were LLM-judged; this build counts deterministically, and the three break down as **1 validator block** + **2 no-evidence** + **0 provider errors**. The eight correct out-of-scope declines are excluded, because declining a question outside the service boundary is intended behaviour. Directional, not a precise percentage improvement.*

Why the unusable rate is the number to look at

System C recorded **zero unsafe actions and twelve unhelpful ones**. It looks excellent on safety and is a poor product, because a system that refuses everything is perfectly safe.

> **Safety is not usefulness**, and measuring only one of them is how you ship the other.

The previous build's own component analysis isolates the cost precisely, because it built System B+ specifically to separate the two additions:

STEP	ADDS	COST IN UNHELPFUL REFUSALS	UNSAFE PREVENTED
B → B+	output validator	+1	–
B+ → C	LLM evidence judge	+6	1

The output validator is effectively free; the judge costs six times that for one prevented case. So this build keeps deterministic output validation and drops the judge — which is that project's own recommendation, not a new one.

Architecture

`validate → safeguarding screen → route → resolve* → prepare* → retrieve → generate → check`

** conditional. Neither is a universal stage.*

Only generation uses a hosted LLM. Routing, safeguarding, resolution, query preparation and validation are deterministic rules. Embeddings run locally on `bge-m3`, so no question leaves the machine to be encoded and there is no per-query embedding cost.

The safeguarding screen is a gate, not a route. It runs first on every message, and when it fires nothing downstream executes: no retrieval, no model call, no validator. That ordering is the single most important decision in the build, because retrieval cannot decline — a deliberately out-of-scope question was measured retrieving at 0.668, above most in-scope ones.

Two output contracts, chosen by provenance

	GROUNDED	CONVERSATIONAL
Used for	factual, access	greetings, support, her ambitions
Safe because	every claim carries a citation	it makes no claim at all
Blocked on	an uncited claim	a citation marker with no passage

Which contract applies is decided by which path produced the turn, never inferred from whether citations happen to be present.

Detect broadly, escalate narrowly

One safeguarding route, two severities, implemented as one boolean. **Urgent** — force, threat, assault, contraceptive sabotage, self-harm — puts verified contacts in front of her. A **concern** — pressure, conditional consent, *"he keeps asking"* — is recognised as safeguarding and answered as a conversation, with contacts offered rather than pushed. Treating every coercion signal as an emergency both reads as being passed on when she came to talk, and at scale buries the services in cases that were never emergencies. Nothing contacts anyone on her behalf.

Evaluation

Five evaluations over 121 labelled items, each with criteria fixed before the run.

1 · Decision layer — 52 messages

METRIC	RESULT	NOTE
Overall accuracy	52/52	1.000
Safeguarding recall	12/12	1.000
Safeguarding precision	12/12	1.000
Contrast pairs	6/6	same surface form, different path

2 · Retrieval — 31 questions, tagged by Theory-of-Change driver

METRIC	NATURAL	PREPARED	ORACLE
Adequate@5 — a passage that *answers* her	0.880	0.960	1.000
Hit@5	0.926	0.963	0.926
MRR	0.883	0.920	0.864
Agency drivers, mean MRR	0.611	0.750	0.889

Gold labels are source-level, so Hit@5 is an optimistic upper bound and is reported as one. Questions are tagged by the eight behavioural drivers rather than generic RAG categories, because knowledge is one driver of eight: this retriever scored knowledge MRR 1.000 while self-identity sat at 0.417.

3 · Multi-turn — 4 journeys, 23 turns

Forbidden passages **1** → **0**. Before conversation state existed, *"and does it hurt?"* asked after a question about the implant retrieved **female sterilization**, and *"where can I go?"* after a coercion disclosure retrieved **BTL**, which is permanent. Path accuracy 21/23; the two disagreements are disputed labels, ours, left uncorrected and documented.

4 · Mixed-intention messages — 15 messages

Wanted evidence **8** → **9** of 14, unwanted passages **4** → **3**, and a purely emotional control that now retrieves nothing at all. Clause-level retrieval, with no word list deciding which clause is the health question — the similarity scores do that.

5 · Conversation quality — 19 journeys, 38 turns

PROPERTY	RESULT
Grounded turns that cite a source	24/24 · 100%
Reply matches her register	38/38 · 100%
Usable reply, not a refusal	38/38 · 100%
Free of a deferral	38/38 · 100%
Free of machinery talk	37/38 · 97%
Free of a passive standing offer	36/37 · 97%
Gives her somewhere to go next	32/37 · 86%

There is no LLM judge, and warmth is deliberately not measured. Every property above is defined in advance and checkable by reading the text, so the score means the same thing on every run and any row can be verified by hand. Warmth cannot be scored honestly by a machine — a judge asked to try produces false positives on exactly the warm, personal turns that matter most — and naming that boundary is more useful than a number nobody can reproduce.

Getting her to a service

The Theory of Change runs behavioural drivers → intent → **service access** → behaviour change. A system that answers every question well and never gets her to a service has done the easy half.

HER TURN	WHAT SHE GETS
"Where can I get family planning near me?"	the cited answer, plus Marie Stopes and One2One with numbers
"Where can I go for an HIV test?"	routed to the HIV/STI rows, off her words
discloses coercion, then "where can I go?"	the help pathway, **0 model calls**
self-harm risk	crisis lines in front of her, not behind a tap
pressure without force	contacts offered behind the tap, not pushed

Contacts are read from a verified table and **never generated**. No corpus chunk contains a phone number, so a number in generated text came from the model's memory, and the validator treats that as fatal. A row reaches a girl only once it carries a source, a checker and a date.

Scope and limitations

- **The evaluation sets are authored in-house** — 121 labelled messages and questions across four sets, written alongside the system. A rigorous regression harness and a design instrument; testing with real users is the next step.
- **The corpus is weighted toward provider guidance** — 1,326 clinical chunks against 58 written for a young reader, which is what clause-level retrieval and the vocabulary mappings exist to bridge.
- **Asking in Kiswahili costs 0.062 similarity**, measured over five matched pairs. Direct translation is handled; idiomatic Sheng is harder.
- **Continuity sits at 86%**, the newest metric and the one with most headroom.

What was deliberately not built

No LLM evidence judge, no output judge, no turn planner, no model-based query rewriter as a first resort, no conversation summariser, no entity tracker, no per-girl profile, no external monitoring vendor. Observability is a JSONL file holding operational fields only — no message text, no identifier for her — because an event log for a safeguarding product, kept by default, is a surveillance database with a dashboard on it.